science to use niche technology to achieve sustainable impact by translating their ‘ideas and discoveries into affordable products and services’. The programme has been intellectually fortified by the association of over sixty faculty members spanning a wide spectrum of expertise at MIT and the association of over 100 students who are Tata Fellows. It has more than forty active projects in ‘India, Nepal, and sub-Saharan Africa’ and works across sectors like ‘agriculture’, ‘energy’, ‘environment’, ‘health’, ‘housing’ and ‘water’. The alumni of the Tata Center have formed new NGOs to share beneficial outcomes of the research undertaken at MIT with similar organizations and government bodies.

The innovations that lead to the creation of products have been designed to be scalable and highly efficient to survive in the market environment in India. ‘What we want through the collaboration with the institution is to transfer the motivation and the openness of MIT’, where ‘you lose the ability to define a student from a faculty member’ because ‘there’s something there that’s just different’, says Ratan. In this institution, ‘there’s so much interdisciplinary work that happens’, he adds.

To ensure sustained and focused attention on the programmes that can benefit India, Ratan worked directly with MIT’s then president, Susan Hockfield. Since her retirement, he has been working with her successor, Rafael Reif (MIT president from July 2012–December 2022), with whom he met Prime Minister Narendra Modi in January 2016 to explain the objectives of the centre and how they would adapt advanced technologies to India’s local needs. He also ‘apprised the Prime Minister on work being done by MIT in the areas of education, health, water and innovation’.

The MIT Tata Center experiment has proved successful. It has engendered ‘curiosity among MIT faculty and students about how their deep knowledge of science, engineering and policy might be relevant to people in developing countries, and especially India, with much lower incomes, large rural populations, lower levels of education, and growing environmental stress’, comments Reif.

The endowment from Tata ‘provides substantial research funding for faculty who, in response, come forward with promising answers … it has also made the Center an unusually productive source of practical innovation that reaches people in India in the form of new products and services, and scientifically informed policy’, adds Reif.